

Energy_Efficiency Performance Report 2024 (ID: doc-syn-04)

1. Executive Summary & Production Efficiency Analysis

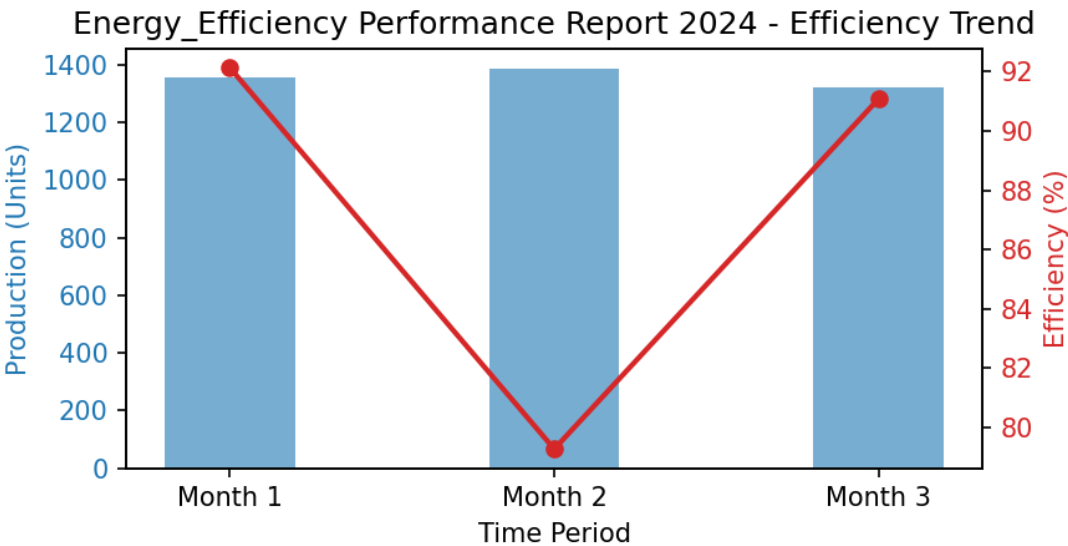
During the operational assessment of Energy_Efficiency Performance Report 2024, production efficiency experienced variance. The primary root cause for downtime in Q2 was unexpected hydraulic failure in Pump-04, whereas in Q4 the downtime was driven by supply chain shortages of raw titanium alloy.

2. Manufacturing Metrics & Output Table

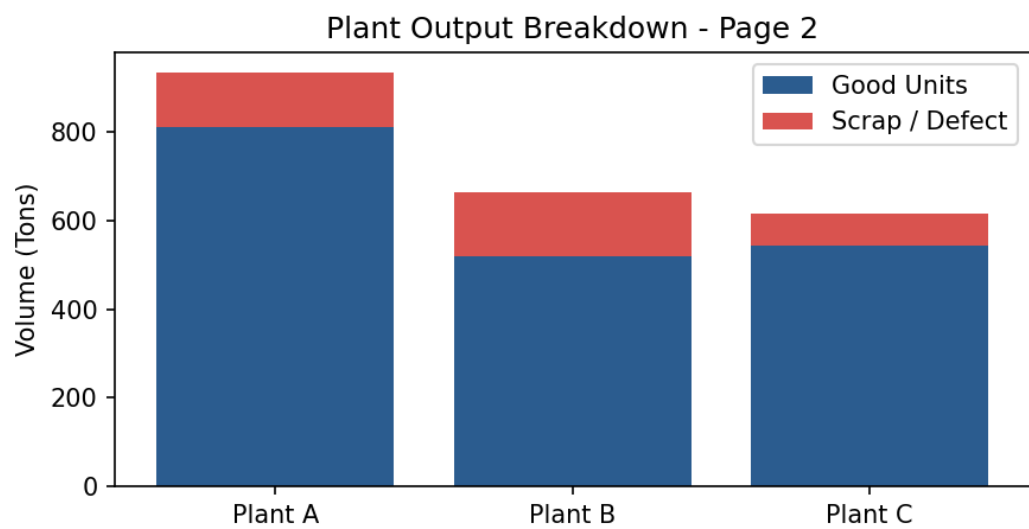
Metric Category	Quarter Baseline	Actual Output*	Efficiency Rate (%)
Line Alpha (Heavy)	1200 Units/Day	1774 Units/Day	86.9%
Line Beta (Light)	800 Units/Day	1241 Units/Day	78.2%
Total Operational Output	2000 Units/Day	3015 Units/Day	82.6%

**Note: Actual output includes unadjusted scrap margin of 3.5% for Units/Day.*

3. Dual-Axis Production vs Efficiency Trend



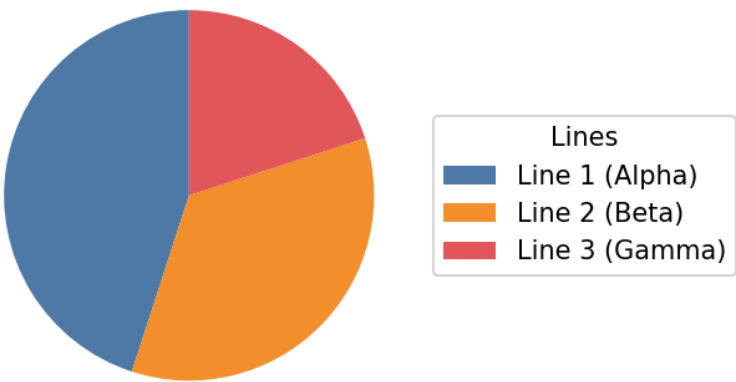
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C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

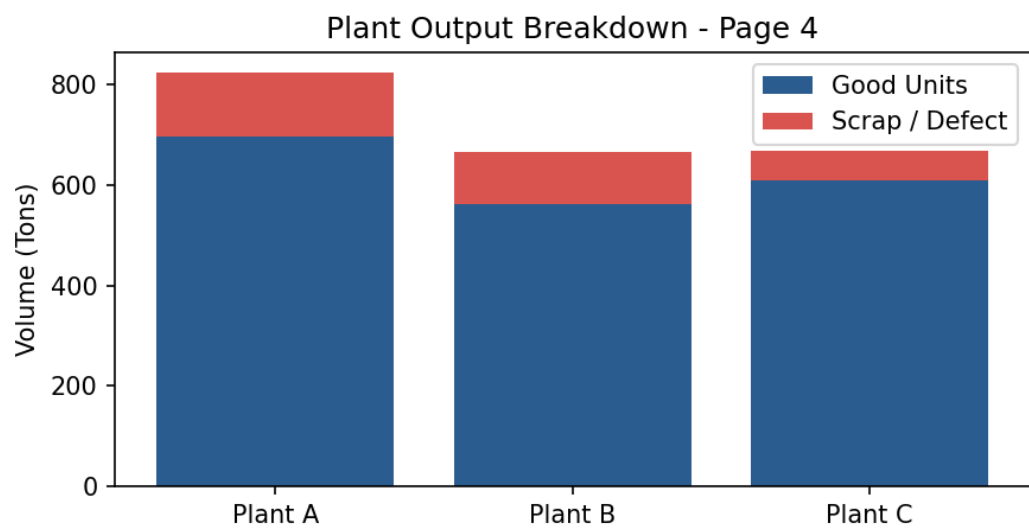
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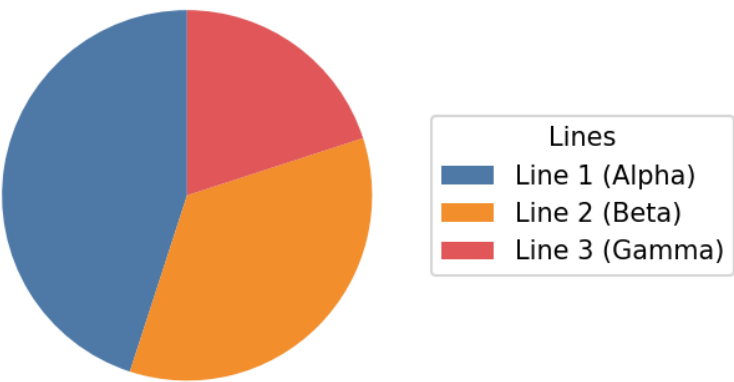
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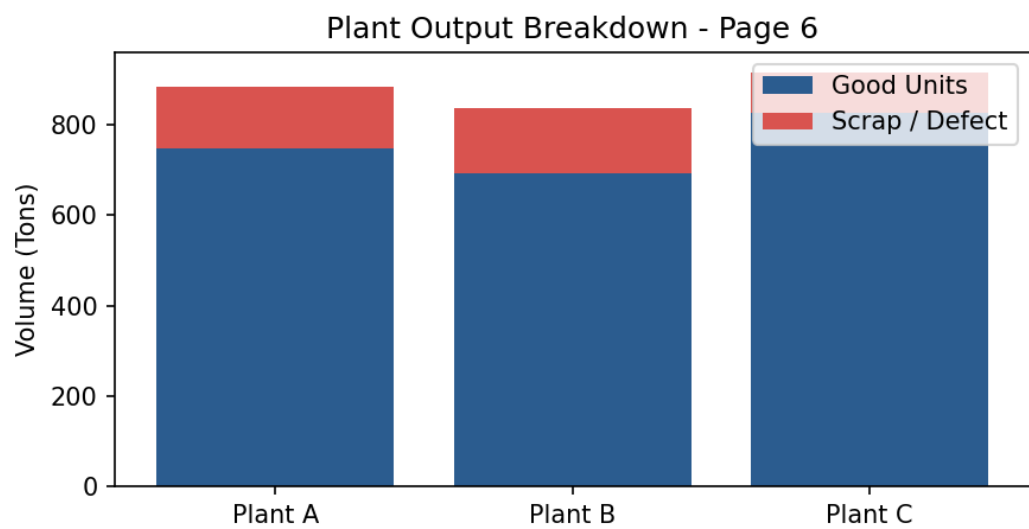
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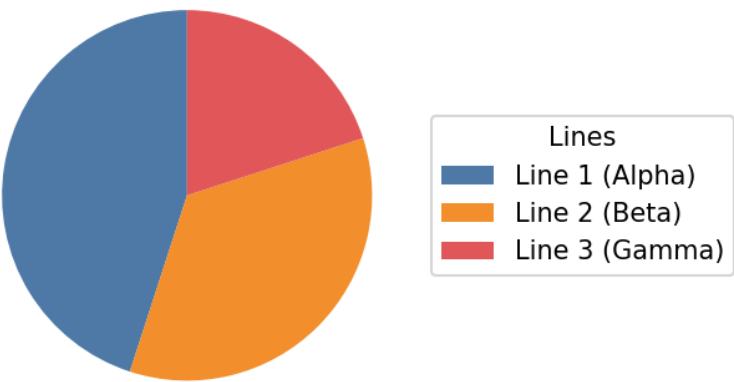
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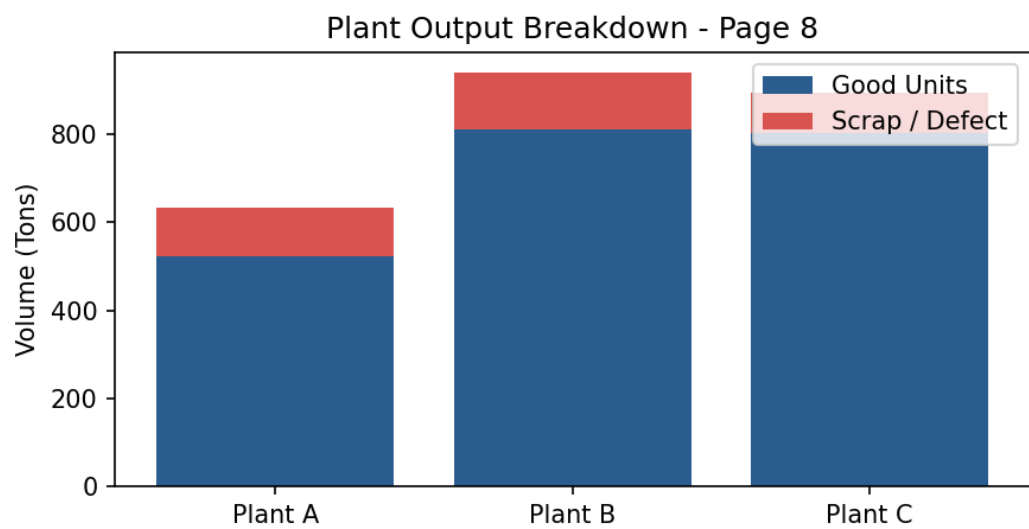
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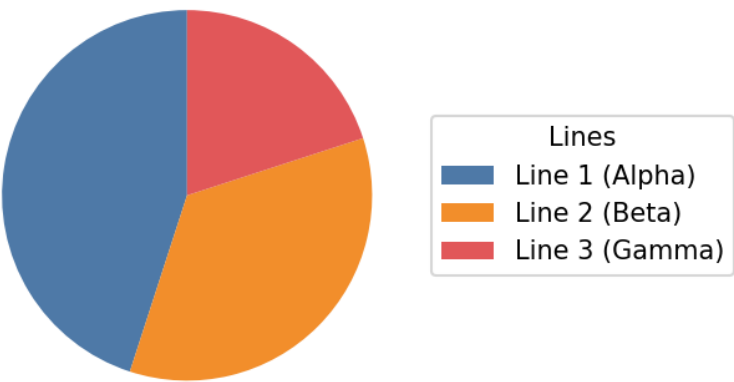
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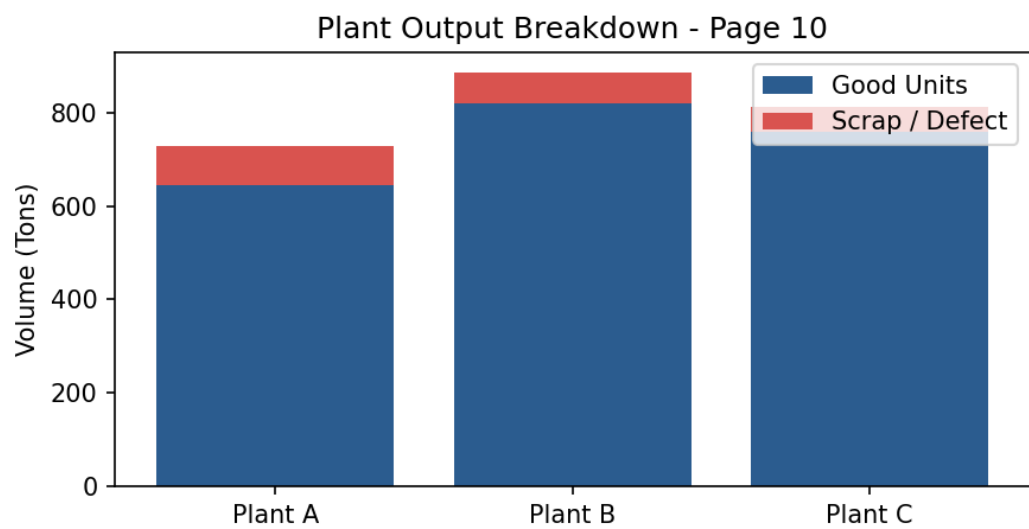
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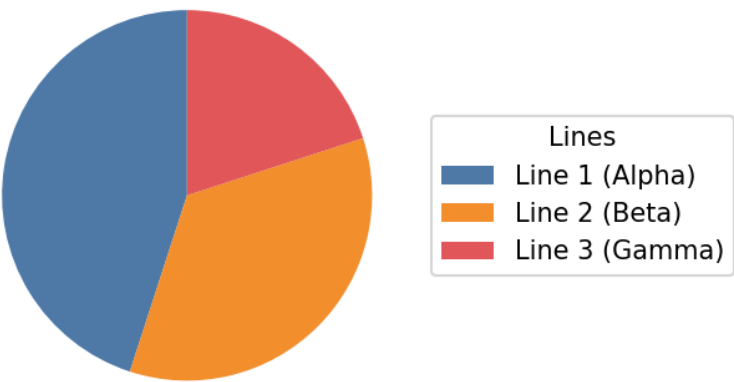
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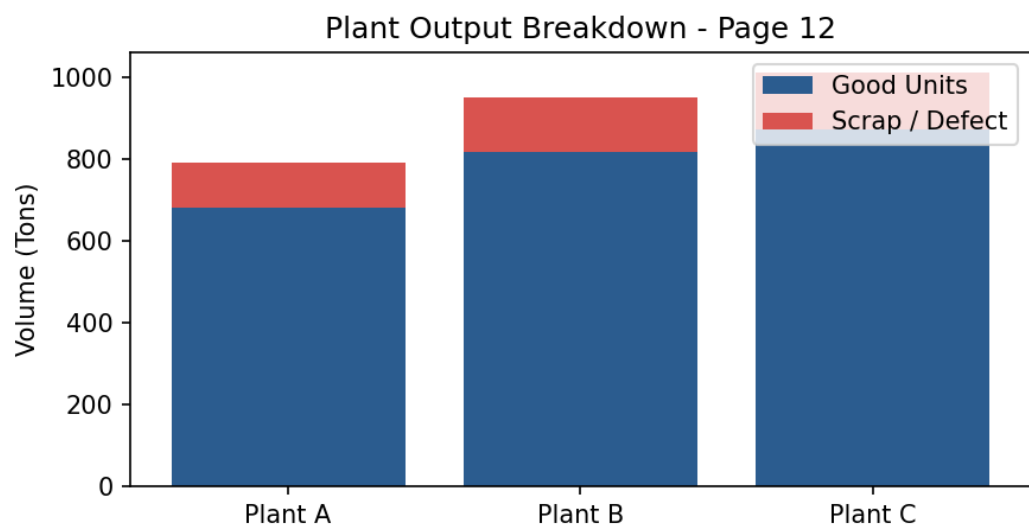
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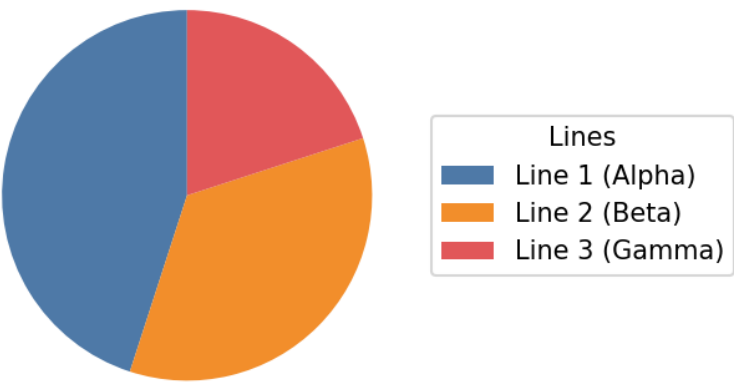
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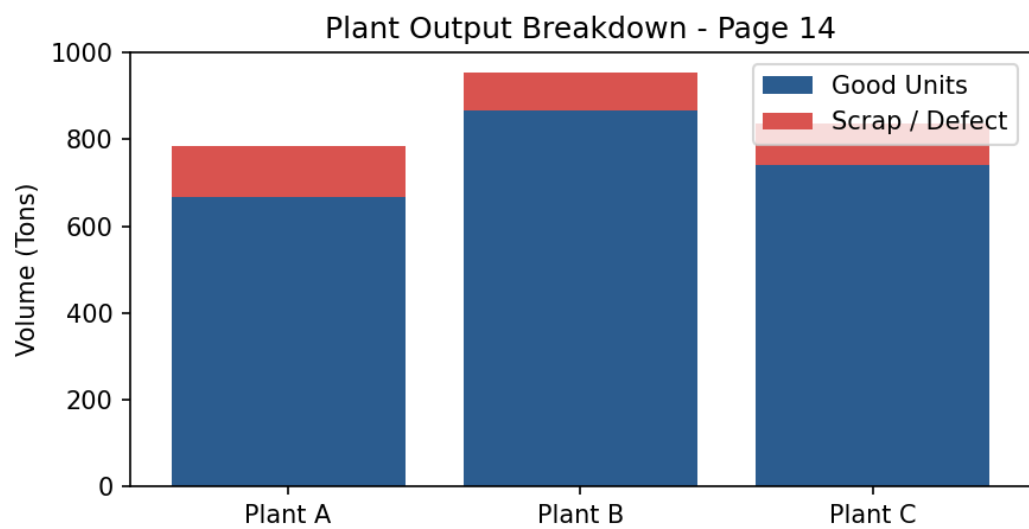
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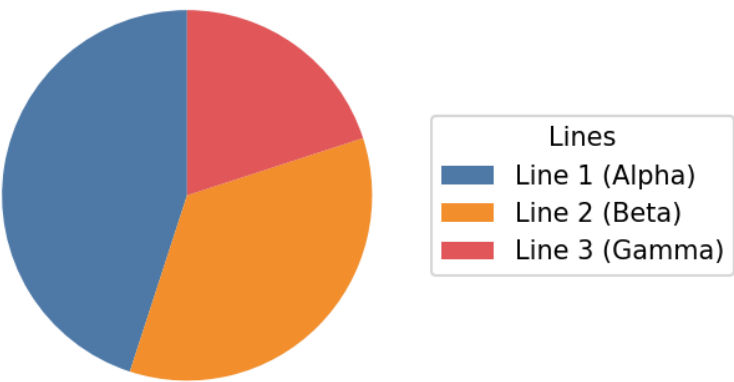
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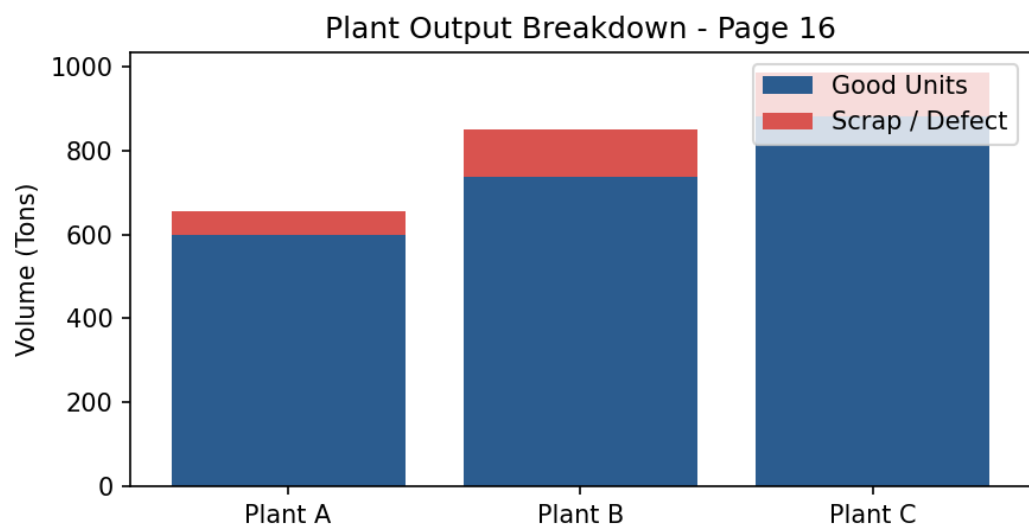
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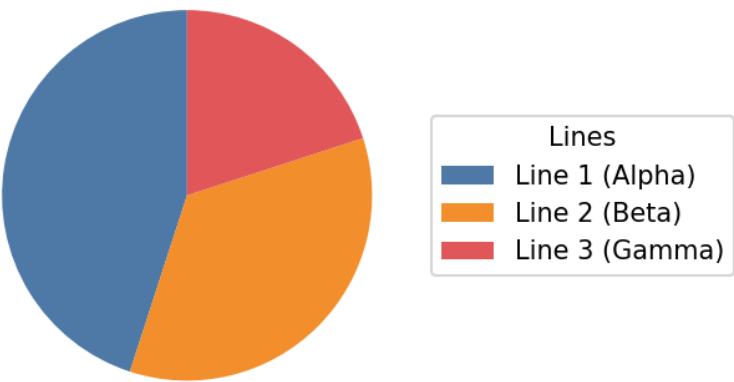
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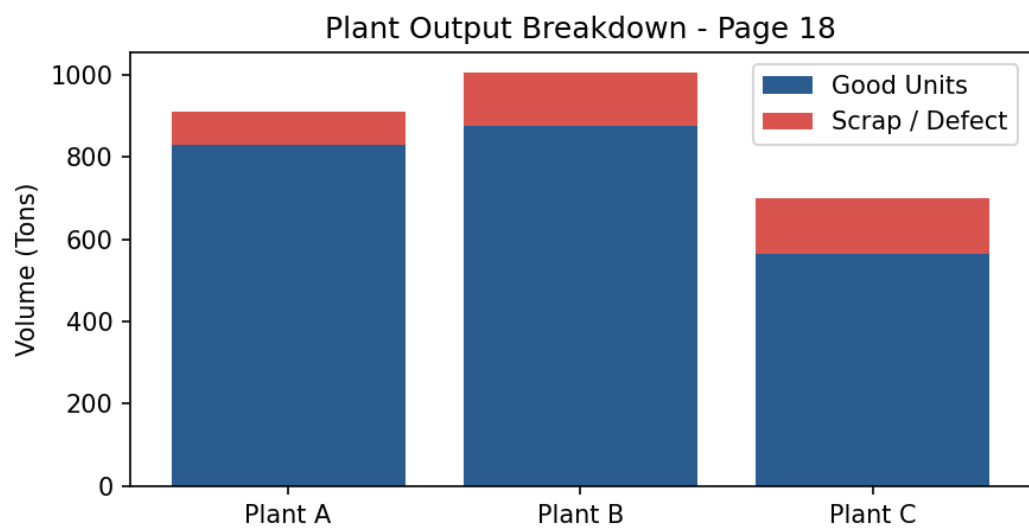
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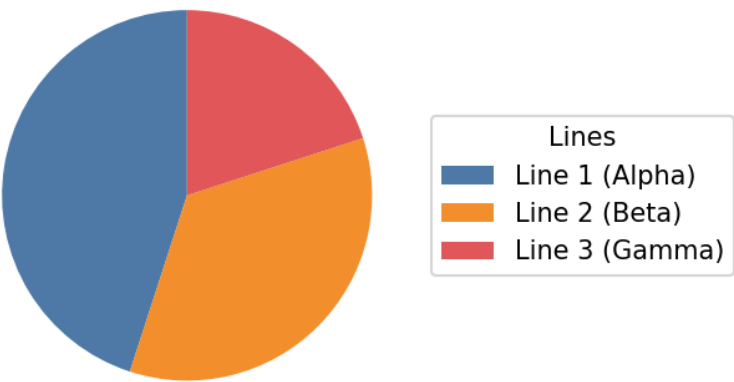
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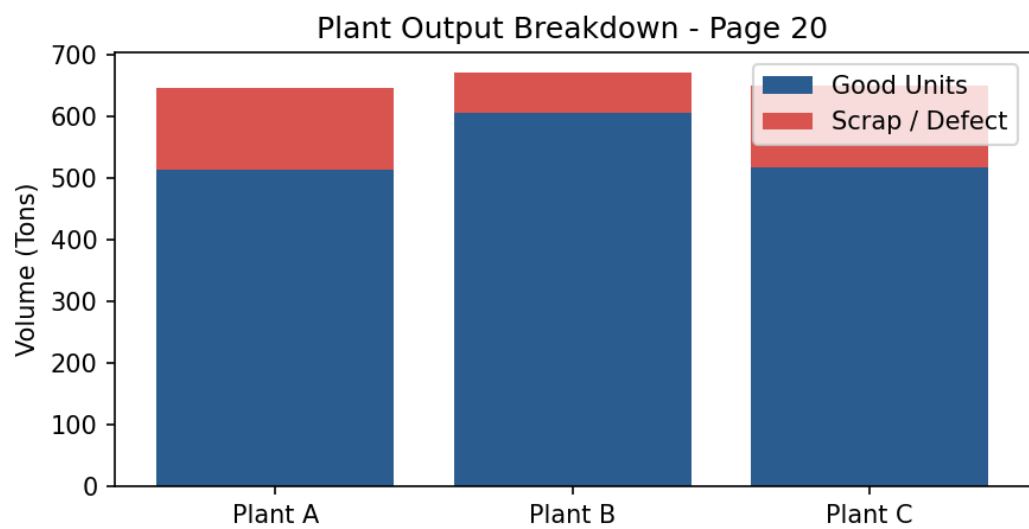
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